

regkit: facilitating manipulation, analysis and visualization of data from Norwegian health and population registers

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Summary

The use of health and administrative registers, often in combination, is an essential component of modern epidemiological research. Among the Nordic countries, in particular, an array of registry sources offers high-quality, broad-coverage data collected across many years (Laugesen et al., 2021). The use of existing data from registers in research circumvents the data collection process, thus making research more cost and time effective (Thygesen & Ersbøll, 2014). Moreover, health and administrative registers offer enormous sample sizes and high representativeness that are much needed, particularly in epidemiological research. Although registers are rich sources of information, pre-processing and working with the large datasets they produce can be challenging and time-consuming – especially for researchers with limited programming experience – and the process might create both unintended variations across projects and errors.

The regkit R package is an open-source toolkit designed to aid researchers in performing efficient and well-documented manipulation, analysis, and visualization of individual-level data from the Norwegian health and population registers. With regkit, we aim to facilitate reproducible descriptive epidemiology based on Norwegian health data, supplemented with sociodemographic information, such as income and education information, from other registries. The package includes functions to validate, filter, and link health (diagnostic) and administrative (sociodemographic) data. For transparency, each function creates a log that documents the function's internal data processing, warnings/errors, and corresponding outputs. Finally, considering the extensive use of registers in epidemiological research, regkit includes functions intended to help users compute common descriptive epidemiology statistics, such as prevalence and incidence rates, and visualize the underlying data.

Statement of need

Due to their characteristics, Nordic registers are highly regarded for their unique potential in current epidemiological research (Jervelund & Montgomery, 2020; Maret-Ouda et al., 2017). In the last decades, epidemiological research in the Nordic countries has showed how advantageous are these registry data (Bakken et al., 2020; Laugesen et al., 2021; Ludvigsson et al., 2016; Viippola et al., 2023). In part, this is possible thanks to the personal identification number, which enables linkage of data for each person from various registries, allowing long-term,

41 multi-dimensional follow-up of individuals in the population. Registry data from national
42 statistical institutes are a widely-used source of auxiliary information in this regard.

43 In Norway, the Norwegian Patient Registry (NPR) is used in a large variety of research projects
44 (Bakken et al., 2020). As of 2025, more than 1000 research papers have been published
45 based on data from the NPR (Norwegian Institute of Public Health, 2024). Statistics Norway
46 (SSB) provides sociodemographic individual-level data on various topics, such as social welfare,
47 education, and income. For instance, between 2021 and 2024, SSB delivered around 900
48 individual-level data assignments to both public authorities and research institutes for analytic
49 and research purposes (Statistics Norway, 2024, 2025). Despite their relatively widespread use
50 in research, health and administrative registers are not designed with research or statistical
51 purposes in mind. This creates numerous potential challenges, inefficiencies, and vulnerabilities
52 in the process of carrying out epidemiological research using linked register data.

53 Considering the wide range of researchers using individual-level registers in Norway, it is highly
54 likely that there are differences in the way researchers pre-process and prepare their data for
55 analysis. Access to register individual-level data is regulated by strict confidentiality laws, which
56 makes “hands-on” training or tutorials hard to access and standardize. The use of proprietary
57 software to manipulate and analyze the data further hinders efforts to ensure reproducibility
58 and transparency across research projects working with the same data (Mathur & Fox, 2023).
59 In this context, we have identified the need for an open-source toolkit to assist researchers
60 working with Norwegian individual-level registry data to prepare, manipulate, and analyze it in
61 a robust, transparent, and reproducible way.

62 State of the field

63 Considering the confidential nature of health and administrative register data, a large proportion
64 of the code used in research is not publicly available. Nonetheless, some few open-source
65 software solutions have been developed to deal with the processing and analysis of Norwegian
66 survey and register data. For instance, phenotools (Hannigan et al., 2021) is an R package
67 that aims to facilitate efficient and reproducible use of survey information from Norwegian
68 cohort data. Similarly, csverse (White, 2025) is a suite of packages focusing on the preparation
69 and analysis of Norwegian disease surveillance and linked register data. Finally, the Norwegian
70 statistics bureau (Statistics Norway) has developed a handful of packages addressing statistical
71 disclosure control, retrieval of publicly available data, classification standards and survey
72 weighting and estimation (Jentof et al., 2026; Langsrud & Bruusgaard, 2026; Langsrud &
73 Lupp, 2025).

74 In summary, these existing projects have showcased the potential of using open-source R
75 packages to assist researchers working with Norwegian survey and register data. The justification
76 for building regkit as a new package, rather than contributing to existing packages, is that
77 currently available software focuses on relatively narrow, well-defined use-cases (e.g., suppression
78 of sensitive information in data tables, disease surveillance) as opposed to broad operations with
79 large, linked individual-level datasets aimed at facilitating descriptive epidemiological analyses.
80 In addition, building a new package has enabled us to create a streamlined framework including
81 only functions relevant to this research area. For instance, it integrates relevant functions
82 originally developed by Statistics Norway to facilitate the creation of realistic simulated data.

83 Software design

84 The package regkit was built following the principles of modularity and flexibility, which
85 increases their possible application in various research projects. For instance, although the
86 package was originally designed for Norwegian data sources, its flexibility may allow for use
87 with other national registries. Similarly, to improve interoperability across projects, regkit
88 imports widely known packages (e.g., ggplot2, dplyr, purrr, etc.) that many users working

89 with register data are likely to have installed already. In addition, packages used in functions
90 only relevant to a specific use case (e.g., `plot_map()`) are listed as *Suggest*, keeping the core
91 installation as lean as possible. The current unit test suite provides good coverage of the main
92 functions, ensuring the reliability of the package.

93 Considering that an average user of `regkit` (social scientist, epidemiologist) may have limited
94 programming experience, we provide user-friendly and educational framework and functions.
95 For instance, one of the first challenges researchers working with population-based registers
96 encounter is how to efficiently manipulate large datasets into smaller and tidier datasets with
97 which they can work analytically. The `regkit` package includes reading and filtering functions
98 (e.g., `read_diag_data()`, `filter_diag_data()`) that support files in parquet format ([Apache
99 Parquet, 2025](#)), which enables to efficiently work with larger-than-memory files in R without
100 requiring deeper knowledge on the inner workings of parquet format objects. Furthermore,
101 most functions automatically generate a log file that records and timestamps the function
102 call, internal data transformations, warnings, errors, and general outputs. These logs can help
103 researchers keep track of and document all manipulation or processing steps applied to their
104 datasets.

105 The package also includes functions that are particularly useful for descriptive epidemiology
106 analyses, such as the computation of prevalence and incidence rates, along with the function
107 `plot_rates()` for visualizing the results. Moreover, there are some specific challenges related
108 to Norwegian registry data that are addressed in the helper functions of `regkit`, such as
109 harmonizing municipality codes and retrieving population counts from SSB's open data.

110 In addition to helping solve practical challenges associated with processing, manipulation, and
111 analysis of Norwegian register data, `regkit` provides “hands-on” guidance on how to efficiently
112 work with individual-level registry data for epidemiological research. The functions in the
113 package are intended to serve as a loose framework that can be adapted by researchers working
114 with similar data and research questions. The package includes a series of vignettes explaining
115 the main functions and real-life examples of descriptive epidemiology. The vignettes and
116 possibility of creating simulated individual-level datasets with the function `simulate_data()`
117 also allow research-groups to use the package as zero-risk training material for new members,
118 and to plan and structure analytic projects prior to obtaining data access.

119 Research impact statement

120 The most prominent use of the `regkit` package to date has been in providing analyses on
121 time trends in autism diagnoses in Norway over recent years, published as part of a national
122 public health report ([Martinez Sanchez et al., 2025](#)). The report received approximately
123 500,000 unique visits throughout 2025, and the analytic component based on work carried
124 out using `regkit` was the most visited section. Core features of this report, which included
125 regional breakdowns and visualization of autism incidence spanning 12 years and prevalence
126 analyses with stratification by demographic characteristics retrieved from Statistics Norway,
127 showcase the key advantages of the software. The package has additionally been used for
128 analyses of time trends in other conditions (e.g., anxiety and depression), with publication of
129 results forthcoming. Moreover, community-readiness has been demonstrated by the acceptance
130 of talks and posters presenting `regkit` at conferences in the fields of epidemiology, autism
131 research, and reproducibility in scientific publishing (e.g. [Martinez Sanchez & Hannigan, 2025](#)).

132 AI Usage Disclosure

133 No generative AI tools were used in the development of this software or the writing of this
134 manuscript.

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